NEET UG 2027: Chemistry Formula Compendium

Module 02: Solutions, Electrochemistry & Chemical Kinetics

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Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Solutions & Colligative Properties (Unit 5)

1.1 Gas Solubility & Raoult's Law

Henry's Law, Raoult's Law & Vapour Pressure

$$p = K_H \cdot x \implies x = \frac{p}{K_H} \quad \text{[Henry's Law: Higher } K_H \text{ at given } T \text{ implies LOWE} \tag{1}$$

$$P_{\text{total}} = P_A + P_B = P_A^\circ x_A + P_B^\circ x_B = P_A^\circ + (P_B^\circ - P_A^\circ)x_B \quad \text{[Raoult's Law for Volatile Binary Liquid Mixtures]} \tag{2}$$

$$y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}, \qquad \frac{1}{P_{\text{total}}} = \frac{y_A}{P_A^\circ} + \frac{y_B}{P_B^\circ} \quad [y_A, y_B \text{ are mole fractions in VAPOUR phase (Dalton} \tag{3}$$

1.2 Colligative Properties & van 't Hoff Factor

The Four Colligative Properties & van 't Hoff Factor (i)

$$\frac{P_A^\circ - P}{P_A^\circ} = i\, x_B = i\left(\frac{n_B}{n_A + n_B}\right) \approx i\left(\frac{w_B M_A}{M_B w_A}\right) \quad \text{[Relative Lowering of Vapour Pressure (RLVP)]} \tag{4}$$

$$\Delta T_b = T_b - T_b^\circ = i\, K_b\, m = i\left(\frac{1000 K_b w_B}{M_B w_A}\right) \quad \text{[Elevation of Boiling Point; } K_b = \frac{R T_b^{\circ 2} M_A}{1000 \Delta H_{\text{vap}}}\text{]} \tag{5}$$

$$\Delta T_f = T_f^\circ - T_f = i\, K_f\, m = i\left(\frac{1000 K_f w_B}{M_B w_A}\right) \quad \text{[Depression of Freezing Point; } K_f = \frac{R T_f^{\circ 2} M_A}{1000 \Delta H_{\text{fus}}}\text{]} \tag{6}$$

$$\Pi = i\, C\, R\, T = i\left(\frac{n_B}{V}\right) RT = i\left(\frac{w_B RT}{M_B V}\right) \quad \text{[Osmotic Pressure; isotonic: } \Pi_1 = \Pi_2\text{]} \tag{7}$$

van 't Hoff Factor (i) Formulations:

$$i = \frac{\text{Experimental / Observed Colligative Property}}{\text{Calculated / Normal Colligative Property}} = \frac{\text{Normal Molar Mass}}{\text{Abnormal / Observed Molar Mass}} \tag{8}$$

$$\alpha_{\text{dissociation}} = \frac{i-1}{n-1} \implies i = 1 + (n-1)\alpha \quad [\text{e.g. for } K_4[Fe(CN)_6] : n = 5 \implies i = 1 + 4\alpha] \tag{9}$$

$$\alpha_{\text{association}} = \frac{1-i}{1-\frac{1}{n}} \implies i = 1 - \left(1 - \frac{1}{n}\right)\alpha \quad [\text{for dimerization } n = 2 : i = 1 - \frac{\alpha}{2}] \tag{10}$$

Solutions Traps: Molality in RLVP & Abnormal Molar Mass

- **RLVP with Molality Shortcut:** For an aqueous solution, relative lowering of vapour pressure can be directly expressed in terms of molality m:

$$\frac{P^\circ - P}{P} = \frac{m \times M_{\text{solvent}}}{1000} \implies \text{For water: } \frac{P^\circ - P}{P} = \frac{18m}{1000}$$

  Notice the denominator is P (solution vapour pressure), NOT P°!
- **Elevation vs Depression Constant:** For water, $K_b = 0.52\,\text{K kg mol}^{-1}$, whereas $K_f = 1.86\,\text{K kg mol}^{-1}$. Freezing point depression is roughly 3.6 times more sensitive than boiling point elevation for the identical molal concentration.

2 Electrochemistry (Unit 7)

2.1 Galvanic Cells, Nernst Equation & Equilibrium

EMF, Nernst Equation & Free Energy Formulations

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = E^\circ_{\text{right}} - E^\circ_{\text{left}} \quad \text{[Standard Reduction Potentials]} \tag{11}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF}\ln Q = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q \quad \text{[Nernst Equation]} \tag{12}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} Q \quad (\text{at } T = 298.15\,\text{K}) \quad [F = 96485\,\text{C mol}^{-1} \approx 96500\,\text{C}] \tag{13}$$

$$E^\circ_{\text{cell}} = \frac{0.0591}{n}\log_{10} K_c \iff \log_{10} K_c = \frac{nE^\circ_{\text{cell}}}{0.0591} \quad [\text{At Equilibrium: } E_{\text{cell}} = 0, \quad Q = K_c] \tag{14}$$

$$\Delta_r G = -nFE_{\text{cell}}, \qquad \Delta_r G^\circ = -nFE^\circ_{\text{cell}} = -2.303RT\log_{10} K_c \quad \text{[Thermodynamic Link]} \tag{15}$$

Concentration Cells (Identical Electrodes, Different Concentrations $C_1 < C_2$):

$$E^\circ_{\text{cell}} = 0 \implies E_{\text{cell}} = \frac{0.0591}{n}\log_{10}\left(\frac{C_2}{C_1}\right) \quad (\text{Spontaneous if } C_2 > C_1) \tag{16}$$

2.2 Electrolytic Conductance & Kohlrausch's Law

Conductivity, Molar Conductivity & Kohlrausch Formulations

$$G = \frac{1}{R} = \frac{A}{\rho L} = \kappa\frac{A}{L} \implies \kappa = G\left(\frac{L}{A}\right) = \frac{G^*}{R} \quad [G^* = \frac{L}{A} = \text{Cell Constant in} \tag{17}$$

$$\Lambda_m = \frac{1000\,\kappa}{M\ [\text{Molarity}]} \quad [\text{S cm}^2\,\text{mol}^{-1}], \qquad \Lambda_{\text{eq}} = \frac{1000\,\kappa}{N\ [\text{Normality}]} = \frac{\Lambda_m}{n\text{-factor}} \tag{18}$$

$$\Lambda_m = \Lambda^\circ_m - A\sqrt{C} \quad [\text{Debye-Hückel-Onsager for S} \tag{19}$$

$$\Lambda^\circ_m(A_xB_y) = x\lambda^\circ_m(A^{y+}) + y\lambda^\circ_m(B^{x-}) \quad [\text{Kohlrausch's Law of Indepe} \tag{20}$$

$$\alpha = \frac{\Lambda_m}{\Lambda^\circ_m}, \qquad K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\Lambda_m^2}{\Lambda^\circ_m(\Lambda^\circ_m - \Lambda_m)}, \qquad S = \frac{1000\,\kappa_{\text{salt}}}{\Lambda^\circ_m} \quad [\text{Sparingly Soluble Salt Solub} \tag{21}$$

2.3 Faraday's Laws of Electrolysis

Faraday's First & Second Laws of Electrolysis

$$w = Z\,I\,t = \left(\frac{E}{96500}\right) I\,t = \left(\frac{M}{n \times 96500}\right) I\,t \qquad \text{[Faraday's First Law; } Q = It\text{]} \qquad (22)$$

$$\frac{w_1}{w_2} = \frac{E_1}{E_2} \quad \text{(for identical charge } Q \text{ passed)} \qquad \text{[Faraday's Second Law]} \qquad (23)$$

Electrochemistry Traps: Intensive E° vs Extensive ΔG°

- **Electrode Potentials are Intensive Properties:** Never multiply E° by stoichiometric coefficients when balancing electrons! For half-reaction multiplied by 2: ΔG° doubles, but E° remains **unchanged**. When combining non-standard half-cells, always combine $\Delta G^\circ = -nFE^\circ$, never add potentials directly!
- **Conductivity vs Molar Conductivity on Dilution:** On dilution: Specific conductivity κ **decreases** (fewer ions per unit volume), but molar conductivity Λ_m **increases** (interionic attractions weaken for strong electrolytes; α increases for weak electrolytes).

3 Chemical Kinetics (Unit 8)

3.1 Rate Laws, Order & Molecularity

Reaction Rates & General Rate Expressions

For general reaction: $aA + bB \rightarrow cC + dD$:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt} \tag{24}$$

$$\text{Differential Rate Law} = k[A]^x[B]^y \implies \text{Overall Order } n = x + y \quad \text{(Order can be fractional or zero)} \tag{25}$$

$$\text{SI Units of Rate Constant } k = (\text{mol}\,\text{L}^{-1})^{1-n}\,\text{s}^{-1} = \text{mol}^{1-n}\,\text{L}^{n-1}\,\text{s}^{-1} \tag{26}$$

3.2 Integrated Rate Equations: Zero, First & n-th Order

Reaction Order	Differential Form	Integrated Equation	Half-Life ($t_{1/2}$)	Completion Time (t_{comp})
Zero Order ($n = 0$)	$-\frac{d[A]}{dt} = k$	$[A]_t = [A]_0 - kt$	$t_{1/2} = \frac{[A]_0}{2k} \propto [A]_0$	$t_{100\%} = \frac{[A]_0}{k} = 2t_{1/2}$
First Order ($n = 1$)	$-\frac{d[A]}{dt} = k[A]$	$k = \frac{2.303}{t}\log_{10}\frac{[A]_0}{[A]_t}$	$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$	$t_{100\%} \rightarrow \infty$ (Never completes)
Second Order ($n = 2$)	$-\frac{d[A]}{dt} = k[A]^2$	$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$	$t_{1/2} = \frac{1}{k[A]_0} \propto \frac{1}{[A]_0}$	$t_{100\%} \rightarrow \infty$
General n-th Order	$-\frac{d[A]}{dt} = k[A]^n$	$\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}} = (n-1)kt$	$t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$	—

Table 1: Systematic Summary of Integrated Rate Formulations and Half-Life Scaling.

First-Order Fractional Lifetimes Shortcuts

For any first-order process:

$$t_{50\%} = 1 \times t_{1/2} = \frac{0.693}{k}$$
$$t_{75\%} = 2 \times t_{1/2} \quad (75\% \text{ consumed, } 25\% \text{ remaining})$$
$$t_{87.5\%} = 3 \times t_{1/2} \quad (87.5\% \text{ consumed, } 12.5\% \text{ remaining})$$
$$t_{99\%} = 2 \times t_{90\%} \approx 6.64 \times t_{1/2} \approx \frac{4.606}{k}$$
$$t_{99.9\%} = 10 \times t_{1/2} \approx 3 \times t_{90\%} = \frac{6.909}{k}$$

3.3 Temperature Dependence: Arrhenius Equation & Catalysis

Arrhenius Equation, Activation Energy & Collision Theory

$$k = A\, e^{-E_a/RT} \implies \ln k = \ln A - \frac{E_a}{RT} \implies \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \tag{27}$$

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \quad [\text{Two Temperatu} \tag{28}$$

$$\text{Slope of } \log_{10} k \text{ vs } \frac{1}{T} = -\frac{E_a}{2.303R}, \qquad \text{Intercept} = \log_{10} A \tag{29}$$

$$\Delta_r H = E_{a,\text{forward}} - E_{a,\text{backward}} \quad [\Delta H < 0 \text{ Exothe} \tag{30}$$

$$\text{Rate} = P \cdot Z_{AB} \cdot e^{-E_a/RT} \quad [P = \text{Steric / Or} \tag{31}$$

Kinetics Traps: Molecularity vs Order & Catalyst Action

- **Molecularity is NEVER Zero or Fractional:** Molecularity is the number of reacting species colliding simultaneously in an elementary step; it is strictly an integer $(1, 2, 3)$. Reactions with molecularity > 3 are statistically rare. Order is experimentally determined and can be 0, fractional, or negative.
- **Catalysts Do NOT Alter Equilibrium State:** A catalyst accelerates both forward and reverse reaction rates by the identical factor by providing an alternate reaction pathway with lower activation energy ($E_a' < E_a$). It does **not** alter $\Delta_r H$, $\Delta_r G^\circ$, or the equilibrium constant K_c.

End of Module 02 Physical Chemistry Formula Handbook (Complete Physical Suite)

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